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UW TCES 330

4 May 2026

Project Phase 1 Report

## Part 1:

Translate the following pseudocode into a sequence of legal instructions; represent machine code in hex.

Pseudocode:

$RF[0A] = D[1B] - D[2A] + D[3C] - D[7E];$

$D[6A] = RF[0A];$

HALT

| Assembly: | Hex machine code: |
|-----------|-------------------|
| LDR 1B A  | 21BA              |
| LDR 2A 1  | 22A1              |
| LDR 3C 2  | 23C2              |
| LDR 7E 3  | 27E3              |
| SUB A 1 A | 4A1A              |
| ADD A 2 A | 3A2A              |
| SUB A 3 A | 4A3A              |
| STR A 6A  | 1A6A              |
| HLT       | 5000              |

## Part 2:

The ALU module was tested in ModelSim and on the DE2 development board. In ModelSim, the ALU was tested using directed tests for all eight required ALU operations, additional edge-case tests for rollover and bitwise behavior, and 1024 randomized test cases using random ALUA, ALUB, and ALUop values. The final simulation completed 1038 total tests with zero errors.

For hardware testing, the ALU was connected to the DE2 board switches, pushbuttons, LEDs, and seven-segment displays through the top-level Part2.sv module. SW[15:0] supplied the 16-bit input value. SW[16] selected which ALU input register was loaded: ALUA or ALUB. SW[17] acted as the load enable. On the rising edge of CLOCK\_50, the selected register loaded the value from SW[15:0].

The stored ALU input values were displayed in hexadecimal on the seven-segment displays. ALUA was shown on HEX7-HEX4, and ALUB was shown on HEX3-HEX0. The ALU

operation was selected using the KEY inputs, with the selected ALUop value displayed on LEDG[3:0]. The 16-bit ALU result was displayed on LEDR[15:0].

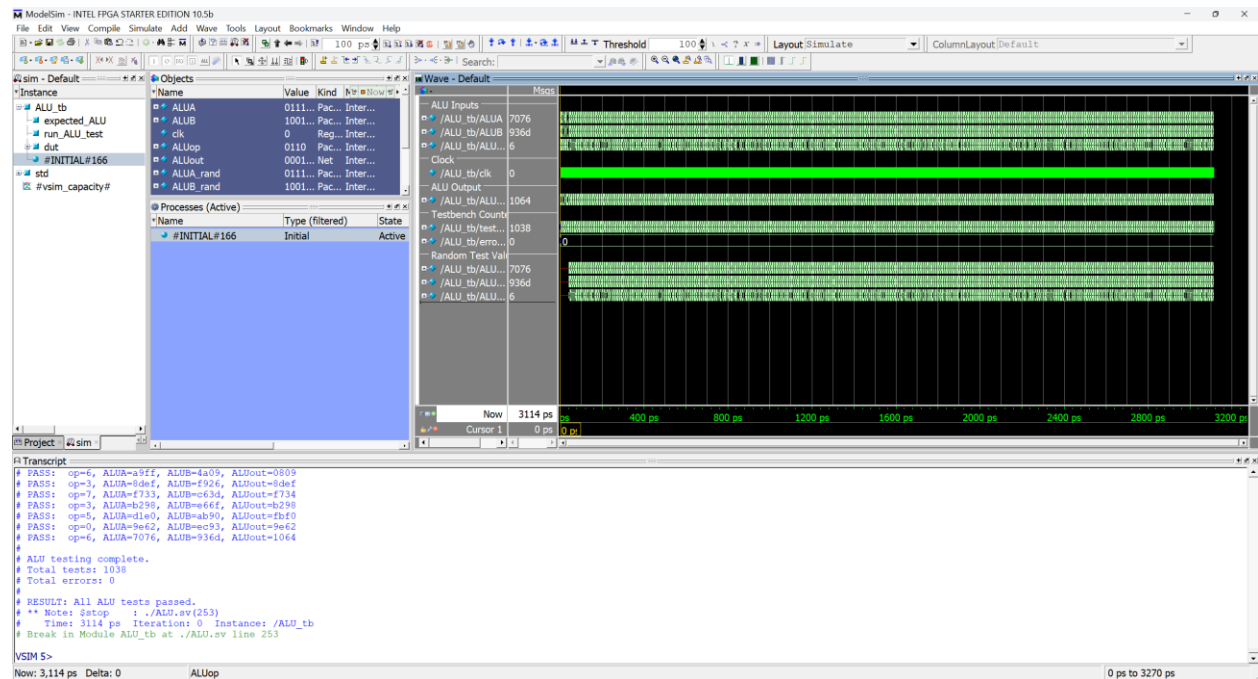


Figure 1 ModelSim Simulation John R. Teal Jr. University of Washington 4 May 2026

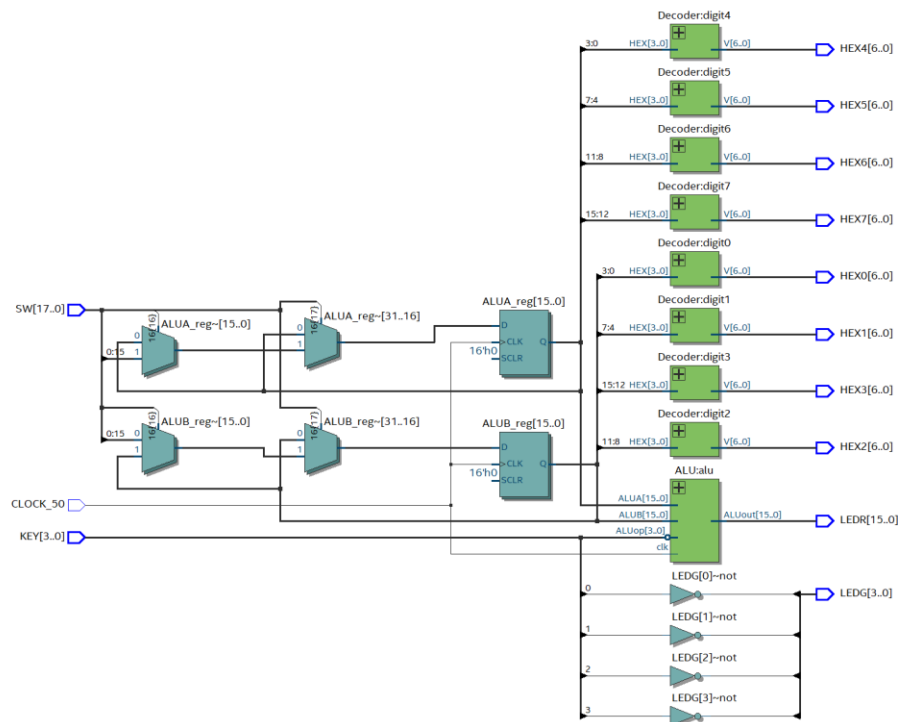


Figure 2 Quartus Circuit John R. Teal Jr. 4 May 2026